Awaiting Accreditation

OCR Level 3 Advanced Subsidiary GCE in Chemistry B (Salters) (H033)

Specification

Version 1: First assessment 2016

This draft qualification has not yet been accredited by Ofqual. It is published to enable teachers to have early sight of our proposed approach to AS Level in Chemistry B (Salters). Further changes may be required and no assurance can be given at this time that the proposed qualification will be made available in its current form, or that it will be accredited in time for first teaching in 2015 and first award in 2016.

Contents

1 Why choose an OCR AS Level in Chemistry B (Salters)?

1a. Why choose an OCR qualification?

Choose OCR and you've got the reassurance that you're working with one of the UK's leading exam boards. Our new AS Level in Chemistry B (Salters) course has been developed in consultation with teachers, employers and higher education to provide students with a qualification that's relevant to them and meets their needs.

We're part of the Cambridge Assessment Group, Europe's largest assessment agency and a department of the University of Cambridge. Cambridge Assessment plays a leading role in developing and delivering assessments throughout the world, operating in over 150 countries.

We work with a range of education providers, including schools, colleges, workplaces and other institutions in both the public and private sectors. Over 13,000 centres choose our A levels, GCSEs and vocational qualifications including Cambridge Nationals, Cambridge Technicals and Cambridge Progression.

Our Specifications

We believe in developing specifications that help you bring the subject to life and inspire your students to achieve more.

We've created teacher-friendly specifications based on extensive research and engagement with the teaching community. They're designed to be straightforward and accessible so that you can tailor the delivery of the course to suit your needs. We aim to encourage learners to become responsible for their own learning, confident in discussing ideas, innovative and engaged.

We provide a range of support services designed to help you at every stage, from preparation through to the delivery of our specifications. This includes:

- A wide range of high-quality creative resources including:
  - delivery guides
  - transition guides
  - topic exploration packs
  - lesson elements
  - …and much more.
- Access to Subject Specialists to support you through the transition and throughout the lifetime of the specifications.
- CPD/Training for teachers including face-to-face events to introduce the qualifications and prepare you for first teaching.
- Active Results – our free results analysis service to help you review the performance of individual students or whole schools.
- ExamCreator – our new online past papers service that enables you to build your own test papers from past OCR exam questions.

All AS level qualifications offered by OCR are accredited by Ofqual, the Regulator for qualifications offered in England.

1b. Why choose an OCR AS Level in Chemistry B (Salters)?

We appreciate that one size doesn't fit all so we offer two suites of qualifications in each science:

Chemistry A – a content-led approach. A flexible approach where the specification is divided into topics, each covering different key concepts of chemistry. Teaching of practical skills is integrated with the theoretical topics and they're assessed both through written papers and, for A level only, the Practical Endorsement.

Chemistry B (Salters) – a context-led approach. Learners study chemistry in a range of different contexts, conveying the excitement of contemporary chemistry. Ideas are introduced in a spiral way with topics introduced in an early part of the course and reinforced later. The 'B' specification places a particular emphasis on an investigational and problem-solving approach to practical work and is supported by extensive new materials developed by the University of York Science Education Group.

All of our specifications have been developed with subject and teaching experts. We have worked in close consultation with teachers and representatives from Higher Education (HE) with the aim of including up-to-date relevant content within a framework that is interesting to teach and administer within all centres (large and small).

Our new AS Level Chemistry B (Salters) qualification builds on our existing popular course. We've based the redevelopment of our AS level sciences on an understanding of what works well in centres large and small and have updated areas of content and assessment where stakeholders have identified that improvements could be made. We've undertaken a significant amount of consultation through our science forums (which include representatives from learned societies, HE, teaching and industry) and through focus groups with teachers. Our papers and specifications have been trialled in centres during development to make sure they work well for all centres and learners.

The content changes are an evolution of our legacy offering and will be familiar to centres already following our courses, but are also clear and logically laid out for centres new to OCR, with assessment models that are straightforward to administer. We have worked closely with teachers and HE representatives to provide high quality support materials to guide you through the new qualifications.

Aims and learning outcomes

OCR's AS Level in Chemistry B (Salters) specification aims to encourage learners to:

- develop essential knowledge and understanding of different areas of the subject and how they relate to each other
- develop and demonstrate a deep appreciation of the skills, knowledge and understanding of scientific methods
- develop competence and confidence in a variety of practical, mathematical and problem solving skills.
- develop their interest in and enthusiasm for the subject, including developing an interest in further study and careers associated with the subject
- understand how society makes decisions about scientific issues and how the sciences contribute to the success of the economy and society (as exemplified in *How Science Works*').

1c. What are the key features of this specification?

Our Chemistry B (Salters) specification has been designed so learners study chemistry in a range of different contexts, conveying the excitement of contemporary chemistry. The specification relates modern-day applications of chemistry and current research to the concepts needed for the study of chemistry at AS Level.

The specification is structured in a series of teaching modules that allow the concepts to unfold throughout the course. Each module is intended to be taught through a chemical 'storyline'. The storylines address topics such as the use and development of fuels, and the use of metals in a wide range of applications including in medicines.

These storylines provide a structure in which to teach the chemical concepts that form the assessable content of the specification. Each storyline brings together concepts from different areas of chemistry, which allows the interconnections between these areas to become clear. As the course progresses, concepts are revisited and built upon in a range of different contexts.

Additionally, the Chemistry B (Salters) specification is designed to stimulate a wide range of practical work. Most storylines offer multiple opportunities for practical work that will help to illustrate the chemical concepts.

A learner of Chemistry B (Salters) will become familiar with exploring key chemistry ideas in a range of contexts. They are able to link chemical concepts together and develop their understanding behind the chemical content within these contexts. Questions within the assessments will be set in unfamiliar contexts, but due to the experience of learning in a range of contexts learners will be comfortable in the application of their chemical knowledge.

The specification:

- has ideas that are introduced within a spiral curriculum structure – topics introduced in an early part of the course and reinforced later
- is laid out clearly in a series of teaching modules with Additional guidance added where required to clarify assessment requirements
- is structured to allow the teaching modules to be taught through chemical 'storylines' that link the specification content with a wide range of contexts
- is co-teachable with the A level
- embeds practical requirements within the teaching modules. Whilst the Practical Endorsement is not part of AS Level in Chemistry B (Salters), opportunities for carrying out activities that would count towards the Practical Endorsement are indicated throughout the specification, in the Additional guidance column, by use of **PAG**, refer to the A level specification, Section 5h, for Practical Endorsement requirements
- exemplifies the mathematical requirements of the course (see Section 5e)
- highlights opportunities for the introduction of key mathematical requirements (see Section 5e and the additional guidance column for each module) into your teaching
- identifies, within the Additional guidance how the skills, knowledge and understanding of How Science Works (HSW) can be incorporated within teaching.

The Chemistry B (Salters) course is fully supported by a dedicated support package written and developed by the University of York Science Education Group, in collaboration with OCR and with sponsorship from the Salters' Institute of industrial chemistry.

Teacher support

The extensive support offered alongside this specification includes:

- **delivery guides –** providing information on assessed content, the associated conceptual development and contextual approaches to delivery
- **transition guides –** identifying the levels of demand and progression for different key stages for a particular topic and going on to provide links to high quality resources and 'checkpoint tasks' to assist teachers in identifying learners 'ready for progression'
- **lesson elements –** written by experts, providing all the materials necessary to deliver creative classroom activities
- **Active Results** (see Section 1a)
- **ExamCreator** (see Section 1a)
- **mock examinations service –** a free service offering a practice question paper and mark scheme (downloadable from a secure location).

Along with:

- Subject Specialists within the OCR science team to help with course queries
- teacher training
- *Science Spotlight* (our termly newsletter)
- OCR Science community
- Practical Skills Handbook
- Maths Skills Handbook

1d. How do I find out more information?

Whether new to our specifications, or continuing on from our legacy offerings, you can find more information on our webpages at: www.ocr.org.uk

Visit our subject pages to find out more about the assessment package and resources available to support your teaching. The science team also release a termly newsletter *Science Spotlight*.

Find out more?

Contact the Subject Specialist:
ScienceGCE@ocr.org.uk, 01223 553998.

Join our Science community:
http://www.cpdhub.ocr.org.uk/

Check what CPD events are available:
www.ocr.org.uk

Follow us on Twitter: @ocr_science

2 The specification overview

2a. Overview of AS Level in Chemistry B (Salters) (H033)

Content Overview	Assessment Overview	
Development of practical skills in chemistry **Storylines** • Elements of life • Developing fuels • Elements from the sea • The ozone story • What's in a medicine?	Foundations of chemistry (01)* 70 marks 1 hour 30 minutes written paper	**50%** of total AS level
	Chemistry in depth (02)* 70 marks 1 hour 30 minutes written paper	**50%** of total AS level

Learners must complete both components (01 and 02).

*Both components include synoptic assessment.

2b. Content of AS Level in Chemistry B (Salters) (H033)

The AS Level in Chemistry B (Salters) specification content is divided into two sections (Section 2c and 2d). An overview of the context is provided at the start of each storyline in Section 2d along with a summary of the chemistry it contains. The assessable content is divided into two columns: **Learning outcomes** and **Additional guidance**.

The Learning outcomes in Sections 2c and 2d may all be assessed in the examinations. The Additional guidance column is included to provide further advice on delivery and the expected skills required from learners.

The details of the storyline contexts, where not directly related to the Learning outcomes, do not form part of the assessable content. These contexts are provided as a coherent and engaging teaching sequence, allowing the specification content to be covered in a way that integrates the various aspects of chemistry and relates the subject to modern applications and everyday experience. Learners will be expected to be able to apply their understanding of chemistry to unfamiliar contexts in the assessments.

References to HSW (Section 5d) are included in the guidance to highlight opportunities to encourage a wider understanding of science.

The mathematical requirements in Section 5e are also referenced by the prefix M (in the additional guidance) to link the mathematical skills required for AS Level Chemistry to examples of science content where those mathematical skills could be linked to learning.

Section 2c in the specification content relates to the practical skills learners are expected to gain throughout the course, which are assessed throughout the written examinations.

Practical activities are embedded within the learning outcomes in Section 2d (Storylines). Suggestions for practical work are also highlighted in the additional guidance (italics) to encourage practical activities in the laboratory, enhancing learners' understanding of chemical theory and practical skills.

The specification has been designed to be co-teachable with the A Level in Chemistry B (Salters) qualification. Learners studying the A level study Section 2c and the first five teaching modules (in Section 2d), continuing with the content of the additional five teaching modules and Chemical literacy in year 13. The internally assessed Practical Endorsement skills also form part of the full A Level (see section 2c, 1.2 in the A level specification).

A summary of the content for the AS level course is as follows:

Section 2c – Development of practical skills in chemistry

- Practical skills assessed in a written examination

Section 2d – Storylines

- Elements of life
- Developing fuels
- Elements from the sea
- The ozone story
- What's in a medicine?

2c. Development of practical skills in chemistry

Module 1: Development of practical skills in chemistry

Chemistry is a practical subject and the development of practical skills is fundamental to understanding the nature of chemistry. Chemistry B gives learners many opportunities to develop the fundamental skills needed to collect and analyse empirical data. Skills in planning, implementing, analysing and evaluating, as outlined in 1.1, will be assessed in the written papers.

1.1 Practical skills assessed in a written examination

Practical skills are embedded throughout all sections of this specification. Suggestions for practical activities are highlighted within Section 2d (Storylines) of the specification in the additional guidance (italics).

Learners will be required to develop a range of practical skills and experiences throughout the course in preparation for the written examinations.

1.1.1 Planning

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	experimental design, including to solve problems set in a practical context	Including selection of suitable apparatus, equipment and techniques for the proposed experiment. Learners should be able to apply scientific knowledge based on the content of the specification to the practical context. HSW3
(b)	identification of variables that must be controlled, where appropriate	
(c)	evaluation that an experimental method is appropriate to meet the expected outcomes.	HSW6

1.1.2 Implementing

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	how to use a wide range of practical apparatus and techniques correctly	As outlined in the Storylines content of the specification (Section 2d). HSW4
(b)	appropriate units for measurements	*M0.0*

(c)	presenting observations and data in an appropriate format.	HSW8

1.1.3 Analysis

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	processing, analysing and interpreting qualitative and quantitative experimental results	Including reaching valid conclusions, where appropriate. HSW5
(b)	use of appropriate mathematical skills for analysis of quantitative data	Refer to Section 5e for a list of mathematical skills that learners should have acquired competence in as part of the course. HSW3
(c)	appropriate use of significant figures	*M1.1*
(d)	plotting and interpreting suitable graphs from experimental results, including selection and labelling of axes with appropriate scales, quantities and units.	*M3.2*

1.1.4 Evaluation

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	how to evaluate results and draw conclusions	HSW6
(b)	the identification of anomalies in experimental measurements	
(c)	the limitations in experimental procedures	
(d)	precision and accuracy of measurements and data, including margins of error, percentage errors and uncertainties in apparatus	*M1.3*
(e)	the refining of experimental design by suggestion of improvements to the procedures and apparatus.	HSW3

2d. Storylines

The content in this section is supported by the resources produced by the University of York Science Education Group (published by Oxford University Press).

The storylines in this section are structured in order to allow teaching of the assessable content in a series of contexts. The contexts are designed to be engaging, and to illustrate the relevance of chemistry in our daily lives and its role in understanding the world around us. At the beginning of each storyline, an overview is given of the intended context and how it relates to the learning outcomes listed below. These context overviews do not form part of the assessable content.

The learning outcomes and additional guidance in each module form the assessable content of the specification; learners will not be expected to recall aspects of the storyline contexts that are not referred to in the learning outcomes. In the final examinations, learners could be assessed on the content of the specification within **any** appropriate context.

Elements of life (EL)

The Big Bang theory is used to introduce the question of where the elements come from. This leads to discussion of the concepts of atomic structure, nuclear fusion, and the use of mass spectroscopy to determine the relative abundance of isotopes.

Next, looking at how we study the radiation we receive from outer space provides the context for discussion of atomic spectroscopy and electronic structure. A historical approach is used to introduce the periodic table, including the links between electronic structure and physical properties. This is followed by studying some of the molecules found in space, providing the context for introducing bonding and structure and the shapes of molecules.

The storyline then turns to chemistry found closer to home. Ideas about the elements found in the human body and their relative amounts are used to introduce the concept of amount of substance and related calculations. The bodily fluids blood and salt then provide a basis for studying salts; this context also incorporates sea water and uses of salts such as in bath salts, lithium batteries, barium meals, hand warmers and fertilisers. This also provides the context for discussing the chemistry of Group 2 elements, as well as amount of substance calculations involving concentration and acid–base titrations.

The chemical ideas in this module are:

- atomic structure, atomic spectra and electron configurations
- fusion reactions
- mass spectroscopy and isotopes
- the periodic table and Group 2 chemistry
- bonding and the shapes of molecules
- chemical equations and amount of substance (moles)
- ions: formulae, charge density, tests
- titrations and titration calculations.

Elements of life (EL)

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
Formulae, equations and amount of substance		
(a)	atomic number, mass number, isotope, Avogadro constant (N_A), relative isotopic mass, relative atomic mass (A_r), relative formula mass and relative molecular mass (M_r)	*M0.0, M0.1, M0.4*
(b)	**(i)** the concept of amount of substance (moles) and its use to perform calculations involving: masses of substances, empirical and molecular formulae, percentage composition, percentage yields, water of crystallisation **(ii)** the techniques and procedures used in experiments to measure masses of solids	*M0.0, M0.1, M0.2, M1.1, M2.2, M 2.3, M2.4* **PAG1** • *experiments involving reacting masses and moles*
(c)	**(i)** the use of the concept of amount of substance (moles) to perform calculations involving: concentration (including titration calculations and calculations for making and diluting standard solutions) **(ii)** the techniques and procedures used in experiments to measure volumes of solutions; the techniques and procedures used in experiments to prepare a standard solution from a solid or more concentrated solution and in acid–base titrations	*M0.0, M0.1, M0.2, M1.1, M1.2, M1.3, M2.2, M2.3, M2.4* At AS, learners will only be expected to carry out structured titration calculations. **PAG2** • *making up standard solutions and diluting solutions using volumetric apparatus* • *acid–base titrations*
(d)	balanced full and ionic chemical equations, including state symbols	*M0.2*
Atomic structure		
(e)	conventions for representing the distribution of electrons in atomic orbitals; the shapes of s- and p-orbitals	The 'electrons in boxes' model.
(f)	the electronic configuration, using sub-shells and atomic orbitals, of: **(i)** atoms from hydrogen to krypton **(ii)** ions of the s- and p-block of Periods 1 to 4 **(iii)** the outer sub-shell structures of s- and p-block elements of other periods	No explanation required. At AS, the electron configurations of Cr and Cu will **not** be assessed.

(g)	how knowledge of the structure of the atom developed in terms of a succession of gradually more sophisticated models; interpretation of these and other examples of such developing models	To include: • evidence for small dense nucleus (Geiger–Marsden experiment) • the make-up of atoms and ions in terms of protons, neutrons and electrons • evidence for electrons shells [from ionisation energies, **EL(q)**, and spectra, **EL(w)**].
(h)	fusion reactions: lighter nuclei join to give heavier nuclei (under conditions of high temperature and pressure); this is how certain elements are formed	Nuclear equations are required.

Bonding and structure

(i)	chemical bonding in terms of electrostatic forces; simple electron '*dot-and-cross*' diagrams to describe the electron arrangements in ions and covalent and dative covalent bonds	In covalent bonds there is a balance between the repulsive forces between the nuclei and the attractive forces between the nuclei and the electrons.
(j)	the bonding in giant lattice (metallic, ionic, covalent network) and simple molecular structure types; the typical physical properties (melting point, solubility in water, electrical conductivity) characteristic of these structure types	Explanations of physical properties limited to: • electrostatic attractions between molecules are weaker than electrostatic attractions in giant structures • charged particles able to move (electrons in metals; ions in molten or aqueous ionic substances).
(k)	use of the electron pair repulsion principle, based on *'dot-and-cross'* diagrams, to predict, explain and name the shapes of simple molecules (such as $BeCl_2$, BF_3,CH_4, NH_3, H_2O and SF_6) and ions (such as NH_4^+) with up to six outer pairs of electrons (any combination of bonding pairs and lone pairs); assigning bond angles to these structures	*M4.1* No treatment of hybridisation or molecular orbitals is expected but ideas of bond angles being altered by the lone pairs present should be included, for example the bond angles of: CH_4 (109.5°), NH_3 (107°), H_2O (104.5°).

(l)	structures of compounds that have a sodium chloride type lattice	Ionic bonding is the overall attraction in a lattice and is made up of attraction between ions of different charge and repulsion between ions of the same charge.

Inorganic chemistry and the periodic table

(m)	the periodic table as a list of elements in order of atomic (proton) number that groups elements together according to their common properties; using given information, make predictions concerning the properties of an element in a group; the classification of elements into s-, p- and d-blocks	
(n)	periodic trends in the melting points of elements in Periods 2 and 3, in terms of structure and bonding	
(o)	the relationship between the position of an element in the s- or p-block of the periodic table and the charge on its ion; the names and formulae of NO_3^-, SO_4^{2-}, CO_3^{2-}, OH^-, NH_4^+, HCO_3^-, Cu^{2+}, Zn^{2+}, Pb^{2+}, Fe^{2+}, Fe^{3+}; formulae and names for compounds formed between these ions and other given anions and cations	When used without oxidation states, 'nitrate' can be assumed to be NO_3^- and 'sulfate' can be assumed to be SO_4^{2-}.
(p)	a description and comparison of the following properties of the elements and compounds of Mg, Ca, Sr and Ba in Group 2: reactions of the elements with water and oxygen, thermal stability of the carbonates, solubilities of hydroxides and carbonates	• *test-tube or reduced scale reactions involving the elements of Group 2 and their compounds*
(q)	the term *ionisation enthalpy*; equations for the first ionisation of elements; explanation of trends in first ionisation enthalpies for Periods 2 and 3 and groups and the resulting differences in reactivities of s- and p-block metals in terms of their ability to lose electrons	Across a period, outermost electrons in the same shell are being more strongly attracted by more protons (explanation of the small drops mid-Period **not** required). Down a group, electrons are in shells that are further from the nucleus and thus attracted less.

Elements of life (EL)

(r)	charge density of an ion and its relation to the thermal stability of the Group 2 carbonates	Smaller ions with the same charge have higher charge density and thus distort the large carbonate ion, so that the compound decomposes at lower temperature.
(s)	the solubility of compounds formed between the following cations and anions: Li^+, Na^+, K^+, Ca^{2+}, Ba^{2+}, Cu^{2+}, Fe^{2+}, Fe^{3+}, Ag^+, Pb^{2+}, Zn^{2+}, Al^{3+}, NH_4^+, CO_3^{2-}, SO_4^{2-}, Cl^-, Br^-, I^-, OH^- , NO_3^-; colours of any precipitates formed; use of these ions as tests e.g. Ba^{2+} as a test for SO_4^{2-}; a sequence of tests leading to the identification of a salt containing the ions above	Knowledge of the reaction of 3+ cations with CO_3^{2-} is **not** required. **PAG4** • *test-tube or reduced scale experiments involving precipitation reactions of the ions in EL(s) and the sequence of tests leading to identification*

Equilibria (acid–base)

(t)	the terms: *acid*, *base*, *alkali, neutralisation*; techniques and procedures for making soluble salts by reacting acids and bases and insoluble salts by precipitation reactions	Knowledge of the names and formulae of the mineral acids, HCl, HNO_3 and H_2SO_4 will be expected. • *making salts (including percentage yield)*
(u)	the basic nature of the oxides and hydroxides of Group 2 (Mg–Ba)	Description only, including equations, for reactions of Group 2 oxides and hydroxides with water and acids.

Energy and matter

(v)	the electromagnetic spectrum in order of increasing frequency and energy and decreasing wavelength: infrared, visible, ultraviolet	

(w) transitions between electronic energy levels in atoms:

- **(i)** the occurrence of absorption and emission atomic spectra in terms of transition of electrons between electronic energy levels
- **(ii)** the features of these spectra, similarities and differences
- **(iii)** the relationship between the energy emitted or absorbed and the frequency of the line produced in the spectra, $\Delta E = h\nu$
- **(iv)** the relationship between frequency, wavelength and the speed of electromagnetic radiation, $c = \nu\lambda$
- **(v)** flame colours of Li^+, Na^+, K^+, Ca^{2+}, Ba^{2+}, Cu^{2+}

Similarities: both are line spectra; lines in same position for a given element; lines become closer at higher frequencies; series of lines representing transitions to or from a particular energy level. Differences: bright/coloured lines on a black background or black lines on coloured/bright background.

• *flame tests for cations*

Modern analytical techniques

(x) use of data from a mass spectrum to determine relative abundance of isotopes and calculate the relative atomic mass of an element.

M1.2, M3.1

Developing fuels (DF)

The use of fuels in cars provides the main context in this storyline, and is used to initially introduce the basic concept of enthalpy change. Food as ‘fuel’ for the body is then an alternative context in which to discuss quantitative aspects of enthalpy, including practical techniques and enthalpy cycles.

The storyline returns to the constituents of car fuels to introduce hydrocarbons and bond enthalpy, after which cracking provides the background to how petrol is produced.

Alkenes are then introduced in the context of saturated and unsaturated fats found in foods. This is followed by studying the polymerisation of alkenes in the context of synthetic polymers and their uses.

The storyline returns to car fuels to discuss combustion reactions and amount of substance calculations involving gases, shapes of hydrocarbons and isomerism, and the atmospheric pollutants produced in burning fuels. The storyline ends by considering the contribution of hydrogen and biofuels as potential fuels of the future.

The chemical ideas in this module are:

- thermochemistry
- organic chemistry: names and combustion of alkanes, alkenes, alcohols
- heterogeneous catalysis
- reactions of alkenes
- addition polymers
- electrophilic addition
- gas volume calculations
- shapes of organic molecules, σ- and π-bonds
- structural and *E/Z* isomers
- dealing with polluting gases.

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
Formulae, equations and amount of substance		
(a)	the concept of amount of substance in performing calculations involving: volumes of gases (including the ideal gas equation *pV* = *nRT*), balanced chemical equations, enthalpy changes; the techniques and procedures used in experiments to measure volumes of gases	*M0.0,M0.1,M0.4, M1.1, M2.2, M2.3, M2.4* The molar gas volume at room temperature and pressure, RTP (24.0 $dm^3\ mol^{-1}$) and the gas constant *R* (8.314 $J\ mol^{-1}\ K^{-1}$) are given on the *Data Sheet.* **PAG1** • *experiments involving reacting masses, moles and volumes of gases*

Developing fuels (DF)

Bonding and structure

(b)	the bonding in organic compounds in terms of σ- and π-bonds	*M4.2*
(c)	the relation of molecular shape to structural formulae and the use of wedges and dotted lines to represent 3-D shape	*M4.2*

Energetics

(d)	the terms: *exothermic*, *endothermic*, *standard conditions*, *(standard) enthalpy change of reaction* ($\Delta_r H$), *(standard) enthalpy change of combustion* ($\Delta_c H$), (*standard) enthalpy change of formation* ($\Delta_f H$), *(standard) enthalpy change of neutralisation* ($\Delta_{neut} H$)	Enthalpy change of neutralisation is per mole of water formed.
(e)	the term *average bond enthalpy* and the relation of bond enthalpy to the length and strength of a bond; bond-breaking as an endothermic process and bond-making as exothermic; the relation of these processes to the overall enthalpy change for a reaction	*M2.4* Understanding of the meaning of 'average' in this context is required.
(f)	techniques and procedures for measuring the energy transferred when reactions occur in solution (or solids reacting with solutions) or when flammable liquids burn; the calculation of enthalpy changes from experimental results	*M0.0, M1.1, M2.3, M3.1, M3.2* Using the formula: $q = mc\Delta T$ **PAG3** • *experiments to measure the energy transferred when reactions occur in solution or when flammable liquids burn*
(g)	the determination of enthalpy changes of reaction from enthalpy cycles and enthalpy level diagrams based on Hess' law	*M0.0, M2.4* Including via enthalpy changes of formation, combustion and bond enthalpies. A statement of Hess' law is **not** required. • *experiments leading to Hess cycles*

Kinetics

(h)	the terms *catalyst*, *catalysis*, *catalyst poison, heterogeneous*	**See also OZ(g)**.

Developing fuels (DF)

(i) a simple model to explain the function of a heterogeneous catalyst

(j) the term *cracking;* the use of catalysts in cracking processes; techniques and procedures for cracking a hydrocarbon vapour over a heated catalyst

Specific examples of catalysts are **not** required.

• *cracking a hydrocarbon vapour over a heated catalyst and testing the product*

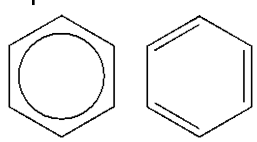
Inorganic chemistry and the table

(k) the origin of atmospheric pollutants from a variety of sources: particulates, unburnt hydrocarbons, CO, CO_2, NO_x, SO_x; the environmental implications and methods of reducing these pollutants

Organic functional groups

(l) the terms *aliphatic*, *aromatic*, *arene*, *saturated, unsaturated, functional group* and *homologous series*

Arenes defined here as compounds containing groups represented as either of:



Unsaturated compounds contain C=C or C≡C.

(m) the nomenclature, general formulae and structural formulae for alkanes, cycloalkanes, alkenes and alcohols (names up to ten carbon atoms)

Organic reactions

(n) balanced equations for the combustion and incomplete combustion (oxidation) of alkanes, cycloalkanes, alkenes and alcohols

(o)	the addition reactions of alkenes with the following, showing the greater reactivity of the C=C bond compared with C–C: **(i)** bromine to give a dibromo compound, including techniques and procedures for testing compounds for unsaturation using bromine water **(ii)** hydrogen bromide to give a bromo compound **(iii)** hydrogen in the presence of a catalyst to give an alkane (Ni with heat and pressure or Pt at room temperature and pressure) **(iv)** water in the presence of a catalyst to give an alcohol (concentrated H_2SO_4, then add water; or steam/H_3PO_4/ heat and pressure)	**PAG7** • *testing compounds for unsaturation using bromine water*

Polymers

(p)	addition polymerisation and the relationship between the structural formula of the addition polymer formed from given monomer(s), and *vice versa*	

Organic mechanisms

(q)	the terms *addition, electrophile, carbocation*; the mechanism of electrophilic addition to alkenes using ‘curly arrows’; how the products obtained when other anions are present can be used to confirm the model of the mechanism	Either a carbocation or a bromonium ion may be shown for bromination.

Isomerism

(r)	structural formulae (full, shortened and skeletal)	
(s)	structural isomerism and structural isomers	*M4.2, M4.3*
(t)	stereoisomerism in terms of lack of free rotation about C=C bonds when the groups on each carbon differ; description and naming as: **(i)** *E/Z* for compounds that have an H on each carbon of C=C **(ii)** *cis/trans* for compounds in which one of the groups on each carbon of C=C is the same	*M4.3* Knowledge of Cahn–Ingold–Prelog (CIP) priority rules will **not** be required.

Sustainability

(u)	the benefits and risks associated with using fossil fuels and alternative fuels (biofuels and hydrogen); making decisions about ensuring a sustainable energy supply.	If comparison with other energy sources is needed, suitable information will be given.

Elements from the sea (ES)

The presence of halide salts in the sea provides the entry to the properties of the halogens and reactions between halide ions. The manufacture of bromine and chlorine then provide the context for discussion of redox chemistry, electrolysis and the nomenclature of inorganic compounds.

The use of chlorine in bleach is used to introduce the concept of equilibrium and calculations of the equilibrium constant, as well as iodine–thiosulfate titrations. This leads into a discussion of the risks and benefits of using chlorine.

Finally, atom economy is introduced through the manufacture of hydrogen chloride and other hydrogen halides. The Deacon process for making HCl provides an opportunity to expand on ideas relating to the position of equilibrium.

The chemical ideas in this teaching module are:

- halogen chemistry
- redox chemistry and electrolysis
- equilibrium
- atom economy.

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
Formulae, equations and amount of substance		
(a)	the concept of amount of substance in performing calculations involving atom economy; the relationship between atom economy and the efficient use of atoms in a reaction	*M0.2*
Redox		
(b)	the explanation (given the necessary information) of the chemical processes occurring during the extraction of the halogens from minerals in the sea	Recall of processes **not** required.
(c)	techniques and procedures in the electrolysis of aqueous solutions; half-equations for the processes occurring at electrodes in electrolysis of molten salts and aqueous solutions: **(i)** formation of oxygen or a halogen or metal ions at the anode **(ii)** formation of hydrogen or a metal at the cathode	*M0.2* Cathode description in aqueous electrolysis: 'Group 1 and 2 and aluminium salts give hydrogen, other metals are plated'. • *electrolysis of aqueous solutions*

Elements from the sea (ES)

(d)	redox reactions of s-, p- and d-block elements and their compounds in terms of electron transfer: **(i)** use of half-equations to represent simple oxidation and reduction reactions **(ii)** the definition of oxidation and reduction as loss and gain of electrons **(iii)** identification of oxidising and reducing agents	*M0.2* Recall of specific reactions is only needed if required elsewhere e.g. **ES(j)**. 'Simple' means not involving acid–base, **see also ES(f)**. • *test-tube or reduced scale redox reactions*
(e)	the oxidation states assigned to and calculated for specified atoms in formulae (including ions) and explanation of which species have been oxidised and which reduced in a redox reaction	
(f)	use of oxidation states to balance redox equations that do not also involve acid–base reactions; techniques and procedures in iodine–thiosulfate titrations	*M0.2* e.g. $3Ca + 2Al^{3+} \rightarrow 3Ca^{2+} + 2Al$ but not: $MnO_4^- + 5Fe^{2+} + 8H^+ \rightarrow Mn^{2+} + 5\ Fe^{3+} + 4H_2O$ • *iodine–thiosulfate titrations*
(g)	use of systematic nomenclature to name and interpret the names of inorganic compounds	e.g. copper(I) sulfide, sodium chlorate(I), lead(II) nitrate(V), potassium manganate(VII) but **not** complex ions.

Inorganic chemistry and the periodic table

(h)	a description of the following physical properties of the halogens: appearance and physical state at room temperature, volatility, solubility in water and organic solvents	Explanation **not** required. **See also OZ(d)**.
(i)	the relative reactivities of the halogens in terms of their ability to gain electrons	• *test-tube or reduced scale reactions involving the halogens and their compounds [related to **(i)** to **(m)**]*
(j)	the details of the redox changes which take place when chlorine, bromine and iodine react with other halide ions, including observations, equations and half-equations	
(k)	the reactions between halide ions (Cl^-, Br^- and I^-) and silver ions (Ag^+) and ionic equations to represent these precipitation reactions, the colours of the precipitates and the solubility of silver halides in ammonia	**PAG4**

Elements from the sea (ES)

(l)	the preparation of HCl; the preparation of HBr and HI by using the halide and phosphoric acid; the action of sulfuric acid on chlorides, bromides and iodides	Details of phosphoric acid (and equations involving it) are **not** required.
(m)	the properties of the hydrogen halides: different thermal stabilities, similar reaction with ammonia and acidity, different reactions with sulfuric acid	Sulfuric acid is reduced to SO_2 by HBr and H_2S by HI.
(n)	the risks associated with the storage and transport of chlorine; uses of chlorine which must be weighed against these risks, including: sterilising water by killing bacteria, bleaching	

Equilibria

(o)	the characteristics of dynamic equilibrium	
(p)	the equilibrium constant, K_c, for a given homogeneous reaction; calculations of the magnitude of K_c using equilibrium concentrations; relation of position of equilibrium to size of K_c, using symbols such as >,<,>>,<<	*M0.1, M2.1* Equilibrium concentrations will be provided. Units will **not** be required.
(q)	the use of K_c to explain the effect of changing concentrations on the position of a homogeneous equilibrium; extension of the ideas of 'opposing change' to the effects of temperature and pressure on equilibrium position.	*M0.3* e.g. 'if a concentration term on the top becomes larger, one on the bottom must also become larger to keep K_c constant, so equilibrium position moves to the left'. • *qualitative experiments involving equilibrium reactions*

The ozone story (OZ)

An initial study of the composition of the atmosphere provides the opportunity to introduce composition by volume calculations for gases.

Discussion of ozone's role as a 'sunscreen' then leads to ideas of the principal types of electromagnetic radiation and their effects on molecules. This introduces a study of radical reactions, reaction kinetics and catalysis, set in the context of the ways in which ozone is made and destroyed in the atmosphere.

A consideration of CFCs and HFCs then provides the introduction to the chemistry of haloalkanes, including nucleophilic substitution, and intermolecular bonding.

The chemical ideas in this module are:

- composition by volume of gases
- the electromagnetic spectrum and the interaction of radiation with matter
- rates of reaction
- radical reactions
- intermolecular bonding
- haloalkanes
- nucleophilic substitution reactions
- the sustainability of the ozone layer.

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
Bonding and structure		
(a)	the term *electronegativity*; qualitative electronegativity trends in the periodic table; use of relative electronegativity values to predict bond polarity in a covalent bond; relation of overall polarity of a molecule to its shape and the polarity of its individual bonds	
(b)	intermolecular bonds: *instantaneous dipole–induced dipole bonds* (including dependence on branching and chain length of organic molecules and M_r), *permanent dipole–permanent dipole bonds*	• *experiments to illustrate the formation of intermolecular bonds (including hydrogen bonds)*
(c)	intermolecular bonds: the formation of *hydrogen bonds* and description of hydrogen bonding, including in water and ice	
(d)	the relative boiling points of substances in terms of intermolecular bonds	This includes an explanation of the boiling points of the halogens.

Kinetics

(e)	the term *activation enthalpy*; enthalpy profiles	Activation enthalpy is related to the energy that pairs of molecules must possess to react when they collide.
(f)	the effect of concentration and pressure on the rate of a reaction, explained in terms of the collision theory; use of the concept of activation enthalpy and the Boltzmann distribution to explain the qualitative effect of temperature changes and catalysts on rate of reaction; techniques and procedures for qualitative experiments in reaction kinetics including plotting graphs to follow the course of a reaction	*M3.2* • *qualitative experiments on reaction kinetics*
(g)	the role of catalysts in providing alternative routes of lower activation enthalpy	**See also DF(h)**.
(h)	the term *homogeneous catalysis* and the formation of intermediates	For example, the catalytic action of chlorine radicals on the breakdown of ozone.

Inorganic chemistry and the periodic table

(i)	calculations, from given data, of values for composition by volume of a component in a gas mixture measured in percentage concentration and in parts per million (ppm)	*M0.0, M0.1*

Organic functional groups

(j) the recognition of and formulae for examples of members of the following homologous series:
- **(i)** haloalkanes, including systematic nomenclature
- **(ii)** amines

Organic reactions

(k) the characteristic properties of haloalkanes, comparing fluoro-, chloro-, bromo- and iodo-compounds, considering the following aspects:
- **(i)** boiling points (depend on intermolecular bonds)
- **(ii)** nucleophilic substitution with water and hydroxide ions to form alcohols, and with ammonia to form amines

Reaction mechanisms

(l) the terms *substitution* and *nucleophile*

(m)	the use of the S_N2 mechanism as a model to explain nucleophilic substitution reactions of haloalkanes using 'curly arrows' and partial charges	Knowledge of the S_N1 mechanism or of the S_N1 or S_N2 nomenclature is **not** required.
(n)	the possible dependence of the relative reactivities of the haloalkanes on either bond enthalpy or bond polarity and how experimental evidence determines that the bond enthalpy is more important	• *experiments to illustrate the relative reactivity of the haloalkanes*
(o)	homolytic and heterolytic bond fission	
(p)	the formation, nature and reactivity of radicals and: **(i)** explanation of the mechanism of a radical chain reaction involving initiation, propagation and termination **(ii)** the radical mechanism for the reaction of alkanes with halogens **(iii)** use of 'half curly arrows' in radical mechanisms	• *experiments involving an alkane and bromine*
(q)	the chemical basis of the depletion of ozone in the stratosphere due to haloalkanes; the ease of photodissociation of the haloalkanes (fluoroalkanes to iodoalkanes) in terms of bond enthalpy	The formation of halogen atoms and the catalytic role of these atoms (and other radicals) in ozone destruction.

Sustainability

(r)	the formation and destruction of ozone in the stratosphere and troposphere; the effects of ozone in the atmosphere, including: **(i)** ozone's action as a sunscreen in the stratosphere by absorbing high-energy UV (and the effects of such UV, including on human skin) **(ii)** the polluting effects of ozone in the troposphere, causing problems including photochemical smog	

Energy and matter

(s)	the principal radiations of the Earth and the Sun in terms of the following regions of the electromagnetic spectrum: infrared, visible, ultraviolet	
(t)	the effect of UV and visible radiation promoting electrons to higher energy levels, sometimes causing bond breaking	
(u)	calculation of values for frequency, wavelength and energy of electromagnetic radiation from given data.	

What's in a medicine? (WM)

A consideration of medicines from nature focuses on aspirin. The chemistry of the –OH group is introduced through reactions of salicin and salicylic acid, beginning with alcohols and continuing with phenols.

The discussion of chemical tests for alcohols and phenols leads to the introduction of IR and mass spectrometry as more powerful methods for identifying substances.

The storyline concludes by examining the synthesis of aspirin to illustrate organic preparative techniques, including a look at the principles of green chemistry.

The chemical ideas in this module are:

- the chemistry of the –OH group, phenols and alcohols
- carboxylic acids and esters
- mass spectroscopy and IR spectroscopy
- organic synthesis, preparative techniques and thin layer chromatography
- green chemistry.

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
Organic functional groups		
(a)	the formulae of the following homologous series: carboxylic acids, phenols, acid anhydrides, esters, aldehydes, ketones, ethers	Nomenclature of compounds with these functional groups is **not** required for AS.
(b)	primary, secondary and tertiary alcohols in terms of the differences in structures	
Organic reactions		
(c)	the following properties of phenols: **(i)** acidic nature, and their reaction with alkalis but not carbonates (whereas carboxylic acids react with alkalis and carbonates) **(ii)** test with neutral iron(III) chloride solution, to give a purple colouration **(iii)** reaction with acid anhydrides (but not carboxylic acids) to form esters	**PAG7**

(d)	the following reactions of alcohols and two-step syntheses involving these reactions and other organic reactions in the AS specification: **(i)** with carboxylic acids, in the presence of concentrated sulfuric acid or concentrated hydrochloric acid (or with acid anhydrides) to form esters **(ii)** oxidation to carbonyl compounds (aldehydes and ketones) and carboxylic acids with acidified dichromate(VI) solution, including the importance of the condition (reflux or distillation) under which it is done **(iii)** dehydration to form alkenes using heated Al_2O_3 or refluxing with concentrated H_2SO_4 **(iv)** substitution reactions to make haloalkanes	**PAG7** • *experiments involving reactions of the OH group*
(e)	techniques and procedures for making a solid organic product and for purifying it using filtration under reduced pressure and re-crystallisation (including choice of solvent and how impurities are removed); techniques and procedures for melting point determination and thin layer chromatography	**PAG6** • *the synthesis and purification of a solid organic compound e.g. aspirin* • *melting point determination* • *the technique of thin layer chromatography (TLC), location of spots and interpretation*
(f)	techniques and procedures for preparing and purifying a liquid organic product including the use of a separating funnel and of Quickfit or reduced scale apparatus for distillation and heating under reflux	**PAG5** • *experiments to oxidise alcohols* • *preparation of an organic compound involving the process of heating under reflux* • *the principal stages in the purification of an organic liquid product e.g. in the preparation of a chloroalkane*
(g)	the principles of green chemistry in industrial processes	*M0.2* Learners should be able to make suggestions based on (but **not** to quote verbatim) the 12 'principles of green chemistry'. Learners will be expected to analyse and use given information. **See also ES(a) EL(b)**

Reaction mechanisms

(h)	the term *elimination reaction*	Example: alkenes from alcohols.

Modern analytical techniques

(i)	interpretation and prediction of mass spectra: **(i)** the M^+ peak and the molecular mass **(ii)** that other peaks are due to positive ions from fragments **(iii)** the M+1 peak being caused by the presence of ^{13}C	*M3.1* Calculations based on M+1 peak will **not** be required.
(j)	the effect of specific frequencies of infrared radiation making specific bonds in organic molecules vibrate (more); interpretation and prediction of infrared spectra for organic compounds, in terms of the functional group(s) present.	*M3.1* IR absorptions will be given on the *Data Sheet*. For AS, questions will only involve hydroxyl, carbonyl and carboxylic acid groups.

2e. Prior learning and progression

This specification has been developed for learners who wish to continue with a study of chemistry at Level 3 in the National Qualifications Framework (NQF). The AS level specification has been written to provide progression from GCSE Science, GCSE Additional Science, GCSE Further Additional Science or from GCSE Chemistry; achievement at a minimum of grade C (or equivalent) in these qualifications should be seen as the normal requisite for entry to AS Level Chemistry. However, learners who have successfully taken other Level 2 qualifications in Science or Applied Science with appropriate chemistry content may also have acquired sufficient knowledge and understanding to begin the AS Level Chemistry course.

There is no formal requirement for prior knowledge of chemistry for entry onto this qualification.

Other learners without formal qualifications may have acquired sufficient knowledge of chemistry to enable progression onto the course.

Some learners may wish to follow a chemistry course for only one year as an AS, in order to broaden their curriculum, and to develop their interest and understanding of different areas of the subject. Others may follow a co-teachable route, completing the one–year AS course and/or then moving to the two–year A level. For learners wishing to follow an apprenticeship route or those seeking direct entry into chemical science careers, this AS level provides a strong background and progression pathway.

There are a number of Science specifications at OCR. Find out more at www.ocr.org.uk

3 Assessment of OCR AS Level in Chemistry B (Salters)

3a. Forms of assessment

Both externally assessed components (01 and 02) contain some synoptic assessment. Component 02 contains some extended response questions.

Foundations of chemistry (Component 01)

This component is split into two sections and assesses content from all teaching modules. Learners answer all questions. This component is worth 70 marks.

Section A contains multiple choice questions. This section of the paper is worth 20 marks.

Section B includes short answer question styles (structured questions, problem solving, calculations, practical) and extended response questions. This section of the paper is worth 50 marks.

Chemistry in depth (Component 02)

This component assesses content from across all teaching modules. Learners answer all questions. This component is worth 70 marks.

Question styles include short answer (structured questions, problem solving, calculations, practical) and extended response questions, including those marked using Level of Response mark schemes.

3b. Assessment availability

There will be one examination series available each year in June to **all** learners.

This specification will be certificated from the June 2016 examination series onwards.

3c. Assessment objectives (AO)

There are three assessment objectives in OCR's AS Level in Chemistry B (Salters). These are detailed in the table below.

Learners are expected to demonstrate their ability to:

	Assessment Objective
AO1	Demonstrate knowledge and understanding of scientific ideas, processes, techniques and procedures.
AO2	Apply knowledge and understanding of scientific ideas, processes, techniques and procedures: • in a theoretical context • in a practical context • when handling qualitative data • when handling quantitative data.
AO3	Analyse, interpret and evaluate scientific information, ideas and evidence, including in relation to issues, to: • make judgements and reach conclusions • develop and refine practical design and procedures.

AO weightings in AS level

The relationship between the assessment objectives and the components are shown in the following table:

Component	**% of AS Level in Chemistry B (Salters) (H033)**		
	AO1	**AO2**	**AO3**
Foundations of chemistry (H033/01)	22–24	19–21	6–8
Chemistry in depth (H033/02)	13–16	21–23	14–15
Total	**35–40**	**40–44**	**20–23**

3d. Written communication skills: assessment of extended responses

The assessment materials for this qualification provide learners with the opportunity to demonstrate their ability to construct and develop a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The marks for extended responses are integrated into the marking criteria.

Extended response questions are included in both externally assessed components. This includes two questions in Component 02, which will be assessed using questions marked by Level of Response, in which the quality of the extended response is explicitly rewarded. These questions will be clearly identified in the assessment papers.

3e. Synoptic assessment

Synoptic assessment tests the learners' understanding of the connections between different elements of the subject.

Synoptic assessment involves the explicit drawing together of knowledge, understanding and skills learned in different parts of the AS level course. The emphasis of synoptic assessment is to encourage the development of the understanding of the subject as a discipline. Both components within Chemistry B (Salters) contain an element of synoptic assessment.

Synoptic assessment requires learners to make and use connections within and between different areas of chemistry, for example, by:

- applying knowledge and understanding of more than one area to a particular situation or context
- using knowledge and understanding of principles and concepts in planning experimental and investigative work and in the analysis and evaluation of data
- bringing together scientific knowledge and understanding from different areas of the subject and applying them.

3f. Calculating qualification results

A learner's overall qualification grade for AS Level in Chemistry B (Salters) will be calculated by adding together their marks from the two components taken to give their total weighted mark.

This mark will then be compared to the qualification level grade boundaries for the relevant exam series to determine the learner's overall qualification grade.

4 Admin: what you need to know

The information in this section is designed to give an overview of the processes involved in administering this qualification so that you can speak to your exams officer. All of the following processes require you to submit something to OCR by a specific deadline.

More information about these processes, together with the deadlines, can be found in the OCR *Admin Guide and Entry Codes: 14–19 Qualifications*, which can be downloaded from the OCR website: www.ocr.org.uk

4a. Pre-assessment

Estimated entries

Estimated entries are your best projection of the number of learners who will be entered for a qualification in a particular series.

Estimated entries should be submitted to OCR by the specified deadline. They are free and do not commit your centre in any way.

Final entries

Final entries provide OCR with detailed data for each learner, showing each assessment to be taken. It is essential that you use the correct entry code, considering the relevant entry rules.

Final entries must be submitted to OCR by the published deadlines or late entry fees will apply.

All learners taking AS Level in Chemistry B (Salters) must be entered using the entry code H033.

Entry option		**Components**		
Entry code	**Title**	**Code**	**Title**	**Assessment type**
H033	Chemistry B (Salters)	01	Foundations of chemistry	External assessment
		02	Chemistry in depth	External assessment

Estimated grades

An estimated grade is the grade the centre expects a learner to achieve for a qualification. These should be submitted to OCR by the specified deadline.

4b. Accessibility and special consideration

The AS level qualification and subject criteria have been reviewed in order to identify any feature which could disadvantage candidates who share a protected Characteristic as defined by the Equality Act 2010. All reasonable steps have been taken to minimise any such disadvantage.

Reasonable adjustments and access arrangements allow learners with special educational needs, disabilities or temporary injuries to access the assessment and show what they know and can do, without changing the demands of the assessment.

Applications for these should be made before the examination series. Detailed information about eligibility for access arrangements can be found in the JCQ *Access Arrangements and Reasonable Adjustments.*

Special consideration is a post-assessment adjustment to marks or grades to reflect temporary injury, illness or other indisposition at the time the assessment was taken. Detailed information about eligibility for special consideration can be found in the JCQ *A guide to the special consideration process.*

4c. External assessment arrangements

Regulations governing examination arrangements are contained in the JCQ *Instructions for conducting examinations*.

4d. Results and certificates

Grade scale

Advanced Subsidiary qualifications are graded on the scale: A, B, C, D, E, where A is the highest. Learners who fail to reach the minimum standard for E will be Unclassified (U). Only subjects in which grades A to E are attained will be recorded on certificates.

Results

Results are released to centres and learners for information and to allow any queries to be resolved **before** certificates are issued.

Centres will have access to the following results' information for each learner:

- the grade for the qualification
- the raw mark for each component
- the total weighted mark for the qualification.

The following supporting information will be available:

- raw mark grade boundaries for each component
- weighted mark grade boundaries for the qualification.

Until certificates are issued, results are deemed to be provisional and may be subject to amendment. A learner's final results will be recorded on an OCR certificate.

The qualification title will be shown on the certificate as 'OCR Level 3 Advanced Subsidiary GCE in Chemistry B (Salters)'.

4e. Post-results services

A number of post-results services are available:

- **Enquiries about results** – If you are not happy with the outcome of a learner's results, centres may submit an enquiry about results.
- **Missing and incomplete results** – This service should be used if an individual subject result for a learner is missing, or the learner has been omitted entirely from the results supplied.
- **Access to scripts** – Centres can request access to marked scripts.

4f. Malpractice

Any breach of the regulations for the conduct of examinations and coursework may constitute malpractice (which includes maladministration) and must be reported to OCR as soon as it is detected.

Detailed information on malpractice can be found in the *Suspected Malpractice in Examinations and Assessments: Policies and Procedures* published by JCQ.

5 Appendices

5a. Grade descriptors

Details to be confirmed by Ofqual.

5b. Overlap with other qualifications

There is a small degree of overlap between the content of this specification and those for other AS level/A level Sciences.

Examples of overlap include:

Biology

- The ozone story: Climate change.
- What's in a medicine: Chromatography.

Geology

- The ozone story: Climate change, the atmosphere.

Physics

- Elements of life: Atomic structure, Atomic emission spectra.

Science

- Elements of life: Atomic structure.
- Developing fuels: Enthalpy changes, catalysis, the development of renewable alternatives to finite energy resources.
- The ozone story: Climate change, the atmosphere, rates of reaction, catalysis.
- What's in a medicine: Infrared spectroscopy, chromatography.

Data Sheet for Chemistry B (Salters)

GCE Advanced Subsidiary and Advanced Level

Chemistry B (Salters) (H033 / H433)

The information in this sheet is for the use of candidates following Chemistry B (Salters) (H033 / H433).

A copy of this sheet will be provided as an insert within the question paper for each component.
Copies of this sheet may be used for teaching.

General Information

Molar gas volume = 24.0 dm^3 mol^{-1} at RTP

Avogadro constant, $N_A = 6.02 \times 10^{23}$ mol^{-1}

Specific heat capacity of water, $c = 4.18$ J g^{-1} K^{-1}

Planck constant, $h = 6.63 \times 10^{-34}$ J Hz^{-1}

Speed of light in a vacuum, c = 3.00×10^8 m s^{-1}

Ionic product of water, $K_w = 1.00 \times 10^{-14}$ mol^2 dm^{-6} at 298 K

1 tonne = 10^6 g

Arrhenius equation: $k = Ae^{-Ea/RT}$ or $\ln k = -E_a/RT + \ln A$

Gas constant, $R = 8.314$ J mol^{-1} K^{-1}

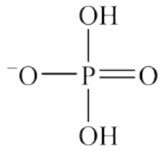
Triplet base codes (codons) for some amino acids used in mRNA

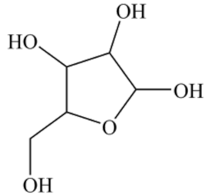
Glycine	GGU
Alanine	GCC
Leucine	CUG
Serine	UCG
Aspartic acid	GAU
Glutamine	CAA
Valine	GUC

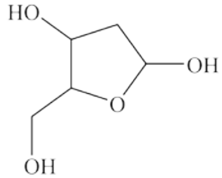
Characteristic infrared absorptions in organic molecules

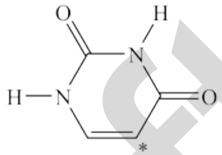
Bond	Location	Wavenumber / cm^{-1}
C–H	Alkanes	2850–2950
	Alkenes, arenes	3000–3100
C–C	Alkanes	750–1100
C=C	Alkenes	1620–1680
aromatic C=C	Arenes	Several peaks in range 1450–1650 (variable)
C=O	Aldehydes	1720–1740
	Ketones	1705–1725
	Carboxylic acids	1700–1725
	Esters	1735–1750
	Amides	1630–1700
	Acyl chlorides and acid anhydrides	1750–1820
C–O	Alcohols, ethers, esters and carboxylic acids	1000–1300
C≡N	Nitriles	2220–2260
C–X	Fluoroalkanes	1000–1350
	Chloroalkanes	600–800
	Bromoalkanes	500–600
O–H	Alcohols, phenols	3200–3600 (broad)
	Carboxylic acids	2500–3300 (broad)
N–H	Primary amines	3300–3500
	Amides	*ca.* 3500

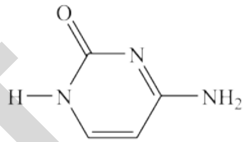
Monomers of DNA and RNA

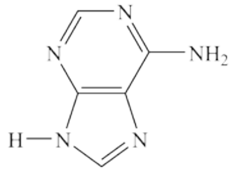

phosphate

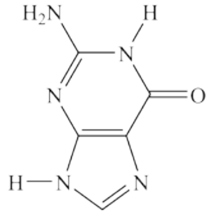

ribose


deoxyribose


uracil


cytosine


adenine

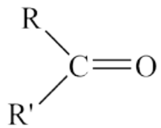
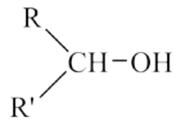

guanine

(thymine has a CH_3 at position *)

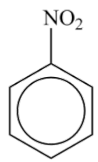
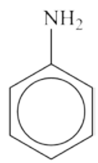
Some useful organic reactions

1 R–Br + CN^- $\longrightarrow$ R–CN + Br^-

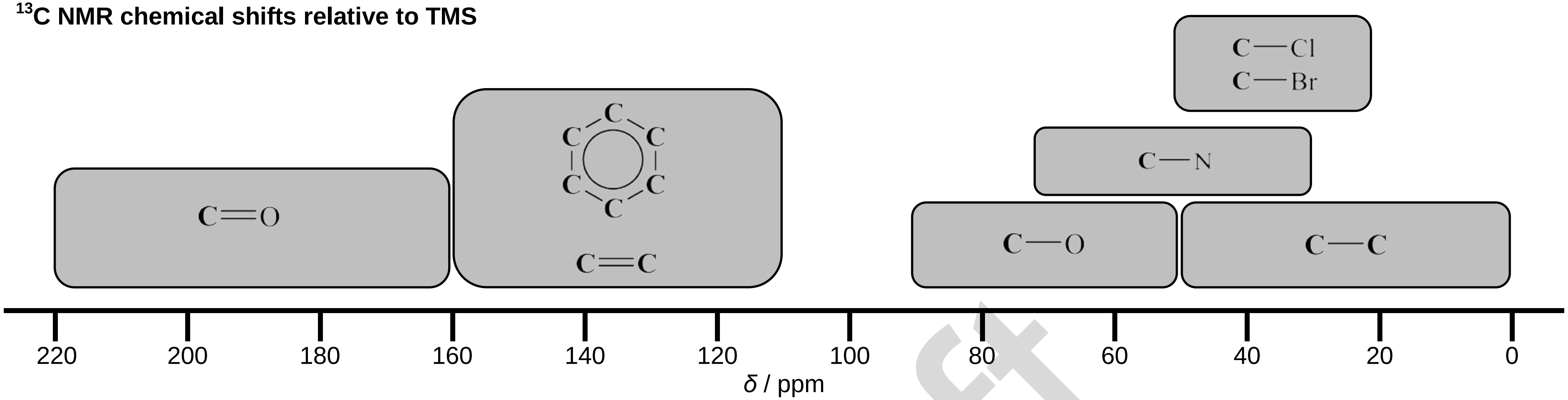
2 R–CN $\xrightarrow[\text{reflux}]{H^+(aq)}$ R–COOH

3  $\xrightarrow{NaBH_4}$ 

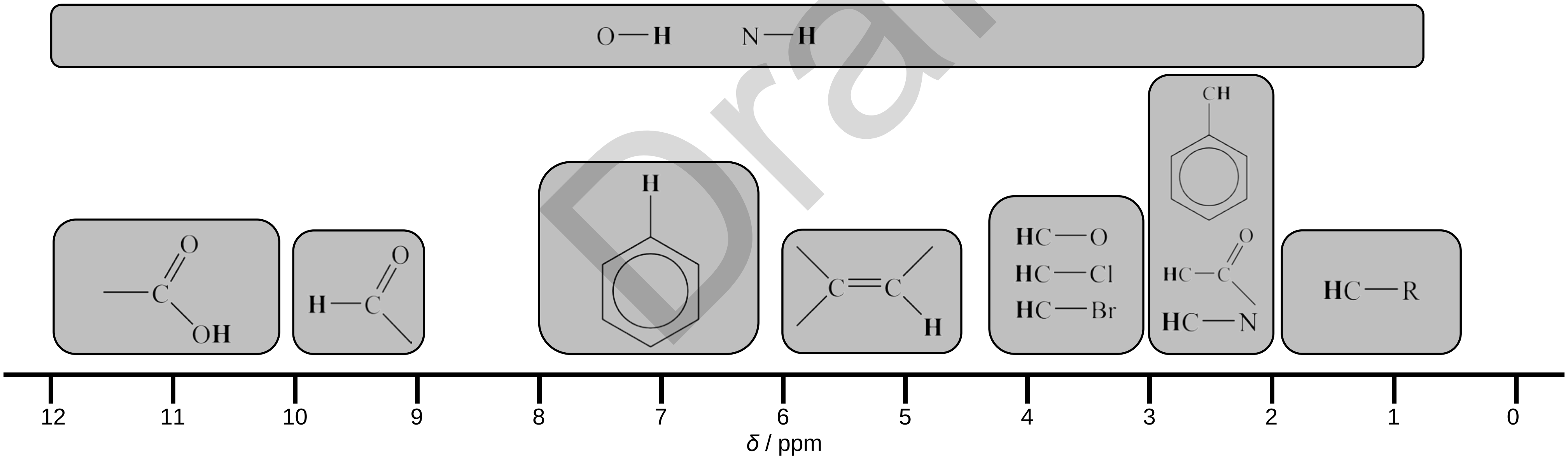
4 R–COOH + $SOCl_2$ $\longrightarrow$ R–COCl + SO_2 + HCl

5  $\xrightarrow[\text{reflux}]{\text{Sn + conc. HCl}}$ 

^{13}C NMR chemical shifts relative to TMS



^{1}H NMR chemical shifts relative to TMS



Chemical shifts are variable and can vary depending on the solvent, concentration and substituents. As a result, shifts may be outside the ranges indicated above.

O**H** and N**H** chemical shifts are very variable and are often broad. Signals are not usually seen as split peaks.

Note that C**H** bonded to 'shifting groups' on either side, e.g. O–C**H**$_2$–C=O, may be shifted more than indicated above.

The Periodic Table of the Elements

Key
atomic number
Symbol
name
relative atomic mass

(1) **1**	(2) **2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	(3) **13**	(4) **14**	(5) **15**	(6) **16**	(7) **17**	(0) **18**
1 **H** hydrogen 1.0																	2 **He** helium 4.0
3 **Li** lithium 6.9	4 **Be** beryllium 9.0											5 **B** boron 10.8	6 **C** carbon 12.0	7 **N** nitrogen 14.0	8 **O** oxygen 16.0	9 **F** fluorine 19.0	10 **Ne** neon 20.2
11 **Na** sodium 23.0	12 **Mg** magnesium 24.3											13 **Al** aluminium 27.0	14 **Si** silicon 28.1	15 **P** phosphorus 31.0	16 **S** sulfur 32.1	17 **Cl** chlorine 35.5	18 **Ar** argon 39.9
19 **K** potassium 39.1	20 **Ca** calcium 40.1	21 **Sc** scandium 45.0	22 **Ti** titanium 47.9	23 **V** vanadium 50.9	24 **Cr** chromium 52.0	25 **Mn** manganese 54.9	26 **Fe** iron 55.8	27 **Co** cobalt 58.9	28 **Ni** nickel 58.7	29 **Cu** copper 63.5	30 **Zn** zinc 65.4	31 **Ga** gallium 69.7	32 **Ge** germanium 72.6	33 **As** arsenic 74.9	34 **Se** selenium 79.0	35 **Br** bromine 79.9	36 **Kr** krypton 83.8
37 **Rb** rubidium 85.5	38 **Sr** strontium 87.6	39 **Y** yttrium 88.9	40 **Zr** zirconium 91.2	41 **Nb** niobium 92.9	42 **Mo** molybdenum 95.9	43 **Tc** technetium	44 **Ru** ruthenium 101.1	45 **Rh** rhodium 102.9	46 **Pd** palladium 106.4	47 **Ag** silver 107.9	48 **Cd** cadmium 112.4	49 **In** indium 114.8	50 **Sn** tin 118.7	51 **Sb** antimony 121.8	52 **Te** tellurium 127.6	53 **I** iodine 126.9	54 **Xe** xenon 131.3
55 **Cs** caesium 132.9	56 **Ba** barium 137.3	57–71 lanthanoids	72 **Hf** hafnium 178.5	73 **Ta** tantalum 180.9	74 **W** tungsten 183.8	75 **Re** rhenium 186.2	76 **Os** osmium 190.2	77 **Ir** iridium 192.2	78 **Pt** platinum 195.1	79 **Au** gold 197.0	80 **Hg** mercury 200.6	81 **Tl** thallium 204.4	82 **Pb** lead 207.2	83 **Bi** bismuth 209.0	84 **Po** polonium	85 **At** astatine	86 **Rn** radon
87 **Fr** francium	88 **Ra** radium	89–103 actinoids	104 **Rf** rutherfordium	105 **Db** dubnium	106 **Sg** seaborgium	107 **Bh** bohrium	108 **Hs** hassium	109 **Mt** meitnerium	110 **Ds** darmstadtium	111 **Rg** roentgenium	112 **Cn** copernicium		114 **Fl** flerovium		116 **Lv** livermorium		

57 **La** lanthanum 138.9	58 **Ce** cerium 140.1	59 **Pr** praseodymium 140.9	60 **Nd** neodymium 144.2	61 **Pm** promethium 144.9	62 **Sm** samarium 150.4	63 **Eu** europium 152.0	64 **Gd** gadolinium 157.2	65 **Tb** terbium 158.9	66 **Dy** dysprosium 162.5	67 **Ho** holmium 164.9	68 **Er** erbium 167.3	69 **Tm** thulium 168.9	70 **Yb** ytterbium 173.0	71 **Lu** lutetium 175.0
89 **Ac** actinium	90 **Th** thorium 232.0	91 **Pa** protactinium	92 **U** uranium 238.1	93 **Np** neptunium	94 **Pu** plutonium	95 **Am** americium	96 **Cm** curium	97 **Bk** berkelium	98 **Cf** californium	99 **Es** einsteinium	100 **Fm** fermium	101 **Md** mendelevium	102 **No** nobelium	103 **Lr** lawrencium

5d. How Science Works (HSW)

How Science Works was conceived as being a wider view of science in context, rather than just straightforward scientific enquiry. It was intended to develop learners as critical and creative thinkers, able to solve problems in a variety of contexts.
Developing ideas and theories to explain the operation of matter and how its composition, structure, properties and changes it undergoes, constitutes the basis of life and all nature**.** *How Science Works* develops the critical analysis and linking of evidence to support or refute ideas and theories. Learners should be aware of the importance that peer review and repeatability have in giving confidence to this evidence.
Learners are expected to understand the variety of sources of data available for critical analysis to provide evidence and the uncertainty involved in its measurement. They should also be able to link that evidence to contexts influenced by culture, politics and ethics.
Understanding *How Science Works* requires an understanding of how scientific evidence can influence ideas and decisions for individuals and society, which is linked to the necessary skills of communication for audience and for purpose with appropriate scientific terminology.

The skills, knowledge and understanding of *How Science Works* (Section 8 of the DfE criteria for science), summarised in the left-hand column below, will underpin the teaching and assessment contexts of the course. The examples given below and within the specification are not exhaustive but give a flavour of opportunities for integrating HSW within the course.

HSW Statement	**Examples of coverage in AS**
HSW1 Use theories, models and ideas to develop scientific explanations.	Developing models to explain atomic structure (**EL**). How elements are formed (**EL**). Using a simple model to explain the function of a catalyst **(DF)**. Using collision theory to explain rates (**OZ**). S_N2 reactions as a model (**OZ**). Using models to explain depletion of ozone (**OZ**).
HSW2 Use knowledge and understanding to pose scientific questions, define scientific problems, present scientific arguments and scientific ideas.	Using *'dot-and-cross'* diagrams to explain shapes of molecules (**EL**). Explaining atomic spectra in terms of electron transitions (**EL**). Calculation of enthalpy changes from experimental techniques (**DF**).
HSW3 Use appropriate methodology, including information and communication technology (ICT), to answer scientific questions and solve scientific problems.	Using oxidation states to balance simple redox equations (**ES**). Using ideas of 'opposing change' to predict the effect of changing conditions on equilibrium position (**ES**).
HSW4 Carry out experimental and investigative activities, including appropriate risk management, in a range of contexts.	Techniques and procedures for making soluble and insoluble salts (**EL**). Use of tests to identify salts (**EL**). Techniques and procedures in iodine and thiosulfate titrations (**ES**). Techniques and procedures in the electrolysis of aqueous solutions (**ES**). Techniques and procedures for making and purifying a solid/liquid organic product (**WM**).

HSW5 Analyse and interpret data to provide evidence, recognising correlations and causal relationships.	Using a graph of first ionisation enthalpy against atomic number to deduce electronic configurations (**EL**). Using experimental observations to explain the reactions between sodium halides and concentrated sulfuric acid (**ES**). Use of data from Mass Spectroscopy to determine relative abundance (**EL**).
HSW6 Evaluate methodology, evidence and data, and resolve conflicting evidence.	Evaluating evidence from the Geiger–Marsden experiment to develop a model for the structure of the atom (**EL**). Evaluating experimental evidence to decide whether the rate of hydrolysis of haloalkanes depends on bond enthalpy or bond polarity (**OZ**).
HSW7 Know that scientific knowledge and understanding develop over time.	Developing models of atomic structure (**EL**).
HSW8 Communicate information and ideas in appropriate ways using appropriate terminology.	Drawing '*dot and cross*' diagrams for simple molecules (**EL**). Explaining the effect of chlorine atoms on the ozone layer (**OZ**).
HSW9 Consider applications and implications of science and evaluate their associated benefits and risks.	Considering the benefits and risks of using fossil fuels and alternative fuels (**DF**). Considering the risks associated with the transport and use of chlorine (**ES**).
HSW10 Consider ethical issues in the treatment of humans, other organisms and the environment.	Considering the environmental implications of atmospheric pollutants (**DF**). Extraction of minerals from the sea (**DF**). Polluting effects of ozone in the troposphere (**OZ**). Considering the use of chlorine in sterilising water (**ES**). The principles of green chemistry (**WM**).
HSW11 Evaluate the role of the scientific community in validating new knowledge and ensuring integrity.	Considering the depletion of ozone in the stratosphere due to haloalkanes (**OZ**).
HSW12 Evaluate the ways in which society uses science to inform decision making.	Evaluating the effect of ozone in the stratosphere (**OZ**).

5e. Mathematical requirements

In order to be able to develop their skills, knowledge and understanding in AS Level Chemistry, learners need to have been taught, and to have acquired competence in, the appropriate areas of mathematics relevant to the subject as indicated in the table of coverage below.

The assessment of quantitative skills will include at least 20% Level 2 (or above) mathematical skills for chemistry (see later for a definition of 'Level 2' mathematics). These skills will be applied in the context of the relevant chemistry.

All mathematical content will be assessed within the lifetime of the specification.

This list of examples is not exhaustive and is not limited to Level 2 examples. These skills could be developed in other areas of specification content from those indicated.

For the mathematical requirements for A Level in Chemistry B (Salters) see the A level specification.

Additional guidance on the assessment of mathematics within chemistry is available on the OCR website.

	Mathematical skill to be assessed	Exemplification of the mathematical skill in the context of AS Level Chemistry (assessment is not limited to the examples below)	Areas of the specification which exemplify the mathematical skill (assessment is not limited to the examples below)
M0 – Arithmetic and numerical computation			
M0.0	Recognise and make use of appropriate units in calculation	Learners may be tested on their ability to: • convert between units e.g. cm^3 to dm^3 as part of volumetric calculations • understand that different units are used in similar topic areas, so that conversions may be necessary e.g. J and kJ.	1.1.2(b) EL(a,b,c), DF(a,f)

	Mathematical skill to be assessed	Exemplification of the mathematical skill in the context of AS Level Chemistry (assessment is not limited to the examples below)	Areas of the specification which exemplify the mathematical skill (assessment is not limited to the examples below)
M0.1	Recognise and use expressions in decimal and ordinary form	Learners may be tested on their ability to: • use an appropriate number of decimal places in calculations • carry out calculations using numbers in standard and ordinary form, e.g. use of Avogadro's constant • convert between numbers in standard and ordinary form • understand that significant figures need retaining when making conversions between standard and ordinary form, e.g. 0.0050 mol dm^{-3} is equivalent to 5.0×10^{-3} mol dm^{-3}.	EL(b), ES(p)
M0.2	Use ratios, fractions and percentages	Learners may be tested on their ability to: • calculate percentage yields • calculate the atom economy of a reaction • construct and/or balance equations using ratios.	EL(b,c,d), ES(a), WM(g)
M0.3	Estimate results	Learners may be tested on their ability to: • use K_c to estimate the position of equilibrium.	ES(q)
M0.4	Use calculators to find and use power functions	Learners may be tested on their ability to: • carry out calculations using the Avogadro constant.	EL(a)
M1 – Handling data			
M1.1	Use an appropriate number of significant figures	Learners may be tested on their ability to: • report calculations to an appropriate number of significant figures given raw data quoted to varying numbers of significant figures • understand that calculated results can only be reported to the limits of the least accurate measurement.	1.1.3(c) EL(b,c), DF(a,f)

	Mathematical skill to be assessed	Exemplification of the mathematical skill in the context of AS Level Chemistry (assessment is not limited to the examples below)	Areas of the specification which exemplify the mathematical skill (assessment is not limited to the examples below)
M1.2	Find arithmetic means	Learners may be tested on their ability to: • calculate weighted means, e.g. calculation of an atomic mass based on supplied isotopic abundances • select appropriate titration data (i.e. identification of outliers) in order to calculate mean titres.	EL(c,x)
M1.3	Identify uncertainties in measurements and use simple techniques to determine uncertainty when data are combined	Learners may be tested on their ability to: • determine uncertainty when two burette readings are used to calculate a titre value.	1.1.4(d), EL(c)
M2 – Algebra			
M2.1	Understand and use the symbols: $=$, $<$, $<<$, $>>$, $>$, $\propto$, $\sim$, $\rightleftharpoons$	No exemplification required.	
M2.2	Change the subject of an equation	Learners may be tested on their ability to: • carry out structured and unstructured mole calculations.	EL(b,c), DF(a)
M2.3	Substitute numerical values into algebraic equations using appropriate units for physical quantities	Learners may be tested on their ability to: • carry out enthalpy change calculations.	EL(b,c), DF(f)
M2.4	Solve algebraic equations	Learners may be tested on their ability to: • carry out Hess' law calculations.	EL(b,c), DF(g)
M3 – Graphs			
M3.1	Translate information between graphical, numerical and algebraic forms	Learners may be tested on their ability to: • interpret and analyse spectra.	EL(x), WM(i,j)

M3.2	Plot two variables from experimental or other data	Learners may be tested on their ability to: • plot graphs from collected or supplied data to follow the course of a reactiondraw lines of best fitextrapolate and interpolate.	1.1.3(d), DF(f), OZ(f)
M4 – Geometry and trigonometry			
M4.1	Use angles and shapes in regular 2-D and 3-D structures	Learners may be tested on their ability to: • predict/identify shapes of and bond angles in molecules with and without a lone pair(s), for example NH_3, CH_4, H_2O etc.	EL(k)
M4.2	Visualise and represent 2-D and 3-D forms including 2-D representations of 3-D objects	Learners may be tested on their ability to: • draw different forms of isomers.	DF(c,s)
M4.3	Understand the symmetry of 2-D and 3-D shapes	Learners may be tested on their ability to: • describe the types of stereoisomerism shown by molecules/ complexes.	DF(s,t)

Definition of Level 2 mathematics

Within AS Level Chemistry, 20% of the marks available within written examinations will be for assessment of mathematics (in the context of chemistry) at a Level 2 standard, or higher. Lower level mathematical skills will still be assessed within examination papers but will not count within the 20% weighting for chemistry.

The following will be counted as Level 2 (or higher) mathematics:

- application and understanding requiring choice of data or equation to be used
- problem solving involving use of mathematics from different areas of maths and decisions about direction to proceed
- questions involving use of A level mathematical content (as of 2012), e.g. use of logarithmic equations.

The following will not be counted as Level 2 mathematics:

- simple substitution with little choice of equation or data
- structured question formats using GCSE mathematics (based on 2012 GCSE mathematics content).

Additional guidance on the assessment of mathematics within chemistry is available on the OCR website as a separate resource, the Maths Skills Handbook.

5f. Health and Safety

In UK law, health and safety is primarily the responsibility of the employer. In a school or college the employer could be a local education authority, the governing body or board of trustees. Employees, (teachers/lecturers, technicians etc), have a legal duty to cooperate with their employer on health and safety matters. Various regulations, but especially the COSHH Regulations 2002 (as amended) and the Management of Health and Safety at Work Regulations 1999, require that before any activity involving a hazardous procedure or harmful micro-organisms is carried out, or hazardous chemicals are used or made, the employer must carry out a risk assessment. A useful summary of the requirements for risk assessment in school or college science can be found at http://www.ase.org.uk/resources/health-and-safety-resources/health-and-safety-risk-assessments/

For members, the CLEAPSS® guide, *PS90, Making and recording risk assessments in school science*[1] offers appropriate advice.

Most education employers have adopted nationally available publications as the basis for their Model Risk Assessments.

Where an employer has adopted model risk assessments an individual school or college then has to review them, to see if there is a need to modify or adapt them in some way to suit the particular conditions of the establishment.

Such adaptations might include a reduced scale of working, deciding that the fume cupboard provision was inadequate or the skills of the candidates were insufficient to attempt particular activities safely. The significant findings of such risk assessment should then be recorded in a "*point of use text*", for example on schemes of work, published teachers guides, work sheets, etc. There is no specific legal requirement that detailed risk assessment forms should be completed for each practical activity, although a minority of employers may require this.

Where project work or investigations, sometimes linked to work-related activities, are included in specifications this may well lead to the use of novel procedures, chemicals or micro-organisms, which are not covered by the employer's model risk assessments. The employer should have given guidance on how to proceed in such cases. Often, for members, it will involve contacting CLEAPSS®.

[1] These, and other CLEAPSS® publications, are on the CLEAPSS® Science Publications website www.cleapss.org.uk . Note that CLEAPSS® publications are only available to members. For more information about CLEAPSS® go to www.cleapss.org.uk.